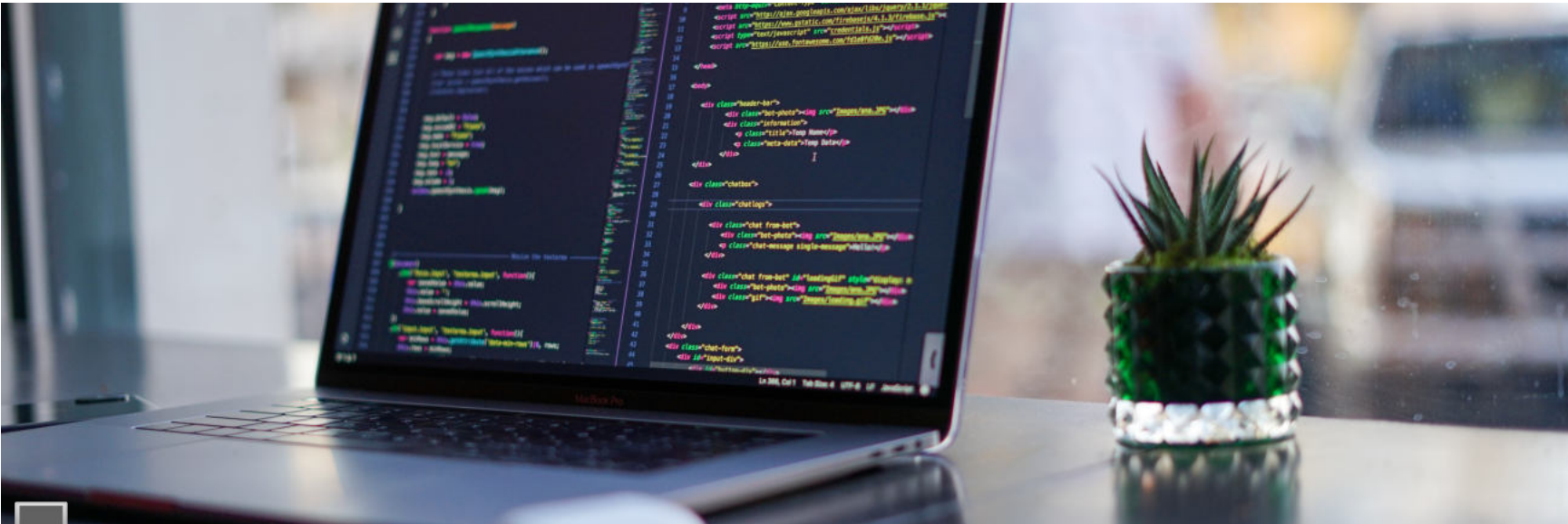




Week 6 Lecture 3

▼ Class	BSCCS2003
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📎 Materials	
# Module #	34
🔵 Type	Lecture
☰ Week #	6

REST APIs

- CRUD
- Variants of listing
- Specialized functions
 - Create a new virtual machine
 - Reboot an existing virtual machine
 - Turn off street lights on a given street
- Formal specifications help others to use

Example #1: Wikipedia

- Open API
- Search for pages
- History of a page
- JSON output

```
curl "https://en.wikipedia.org/w/rest.php/v1/search/page?q=earth&limit=1"
```

Response

```
{
  "pages": [
    {
      "id": 9228,
      "key": "Earth",
      "title": "Earth",
      "excerpt": "<span class=\"searchmatch\">Earth</span> is the third planet from the Sun and the only astronomical object known to harbour and support life. About 29.2% of <span class=\"searchmatch\">Earth's</span> surface is land, with the remainder being the oceans. The average distance between Earth and the Sun is approximately 149.6 million km (93 million miles). Earth is the second largest planet in the Solar System by both diameter and mass, with a surface area of 510 million km² (197 million sq mi) and a mass of 5.972 × 10²⁴ kg (1.315 × 10²⁵ lb). Earth's gravity keeps the atmosphere in place, and its magnetic field helps protect the atmosphere from the solar wind. The atmosphere is composed of nitrogen, oxygen, and trace gases, and is responsible for the greenhouse effect, which keeps the planet warm enough to support life. Earth's internal structure is composed of a solid inner core, a liquid outer core, and a mantle. The mantle is composed of silicate rocks, and the core is composed of iron and nickel. The Earth's rotation is responsible for the day and night cycle, and its axial tilt is responsible for the seasons. The Earth's history is marked by a series of geological events, including the formation of the planet, the development of life, and the extinction of dinosaurs. The Earth's future is uncertain, but it is expected to continue to support life for billions of years."
      "description": "Third planet from the Sun in the Solar System",
      "thumbnail": {
        "mimetype": "image/jpeg",
        "size": null,
        "width": 200,
        "height": 200,
        "duration": null,
        "url": "//upload.wikimedia.org/wikipedia/commons/thumb/9/97/The_Earth_seen_from_Apollo_17.jpg/200px-The_Earth_seen_from_Apollo_17.jpg"
      }
    }
  ]
}
```

Documentation

Schema [\[edit \]](#)

id required integer	Page identifier
key required string	Page title in URL-friendly format
title required string	Page title in reading-friendly format
excerpt required string	For <i>search pages endpoint</i>: A few lines giving a sample of page content with search terms highlighted with <code></code> tags For <i>autocomplete page title endpoint</i>: Page title in reading-friendly format
description required string	Short summary of the page topic based on the corresponding entry on Wikidata or <code>null</code> if no entry exists

Detailed Documentation

Search pages [\[edit \]](#)

Route	/search/page?q=search terms	Content type	application/json
Method	GET	Returns	pages object containing array of search results

Searches wiki page titles and contents for the provided search terms, and returns matching pages.

When using this endpoint on your wiki



This endpoint uses the search engine configured in the `$wgSearchType` configuration setting and returns results in the namespaces configured by `$wgNamespacesToBeSearchedDefault`.

Examples [\[edit \]](#)

curl Python PHP JavaScript

```
# Search English Wikipedia for up to 20 pages containing information about Jupiter
$ curl https://en.wikipedia.org/w/rest.php/v1/search/page?q=jupiter&limit=20
```

Parameters and Response Codes

Parameters [\[edit \]](#)

q required query	Search terms
limit optional query	Maximum number of search results to return, between 1 and 100. Default: 50

Responses [\[edit \]](#)

200	Success: Results found. Returns a <code>pages</code> object containing an array of search results .
200	Success: No results found. Returns a <code>pages</code> object containing an empty array.
400	Query parameter not set. Add <code>q</code> parameter.
400	Invalid limit requested. Set <code>limit</code> parameter to between 1 and 100.
500	Search error